High Altitude Balloons

Launching high-altitude research devices to 100,000ft!

Jacob Boxerman, Claire Cizdziel, Roland List, Nathaniel Rodgers, Claudio Solano,
Daniel Perez, Beruktawit Gebreamlak, Nicolas Garcia, Anika Somaia, Jess Bolar, Eva Sharman

MISSION OBJECTIVE

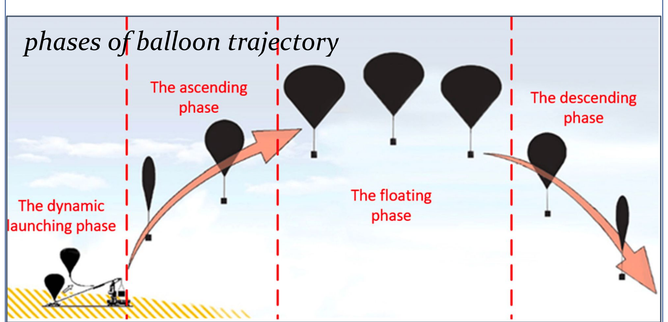
Mission goals:

Balloon: Successfully launch and recover a high altitude balloon.

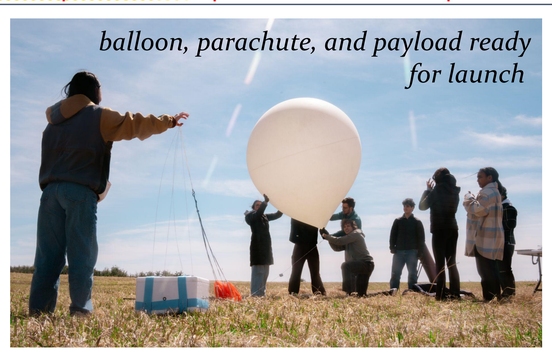
Payload: Design a unique payload system to collect data from sensors up to 100,000ft.

How it works:

Helium lifts the balloon to a desired float altitude. As the balloon rises, the gas inside expands and eventually pops, releasing a parachute as the payload falls to earth



phases of balloon trajectory



balloon, parachute, and payload ready for launch

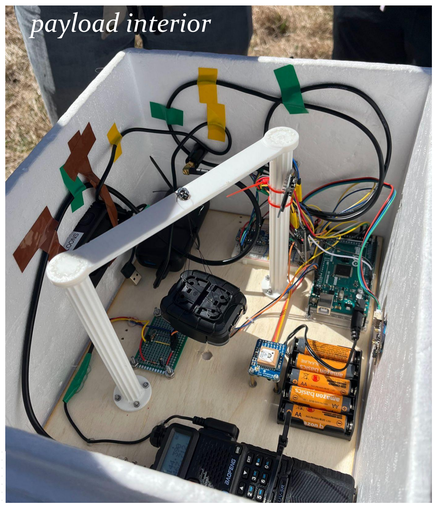
2023-2024 Progress



filling balloon on launch day

Payload:

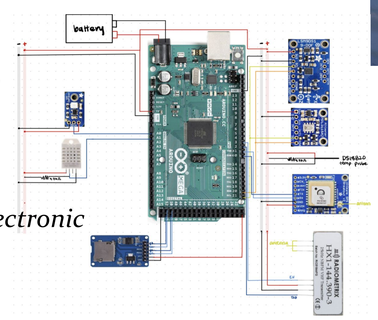
- Houses all tracking, camera, electronics, and power systems in a styrofoam cooler
- Built a custom strapping device to connect the to parachute and balloon to the payload
- Designed and manufactured a custom helium filling mechanism for the balloon
- Used styrofoam box for temperature regulation.



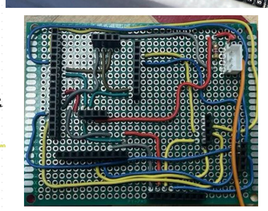
payload interior

Electronics:

- Designed a custom circuit board to combine environmental sensors with power systems, data logging, and camera
- Used Arduino to collect data from sensors,
- Developed code with interface with temperature, UV, altitude, acceleration, and oxygen sensors



preliminary electronic block diagram



custom protoboard with inserts for sensors

Tracking:

- Consisted of a custom-build amateur radio tracker
- Used a handheld transceiver for transmission and Arduino for modulating signals to be received by 'digipeaters' and uploaded to the Internet.
- Backup tracking provided by a commercial SPOT satellite asset tracker.



payload exterior

RESULTS AND FUTURE PLANS

Accomplishments:

Successful launch to 100,000 ft!

Due to high winds, it was challenging to properly fill the balloon. Unfortunately, we underfilled by ~25 cubic feet (only about ⅛ off), giving our balloon a positive lift of 10g instead of 700 g. This led our balloon to float nearly 400 miles off the coast of the US. Wow!



balloon launched!

Next steps:

Second launch in one week! With a new commercial marketing payload

LEARN MORE

- #balloons on slack
- linktr.ee/columbiaspace
- jib2137@columbia.edu
- ctc2156@columbia.edu